

The Transcriptional Coactivator Yes-Associated Protein Drives p73 Gene-Target Specificity in Response to DNA Damage

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Summary

The transcriptional coactivator Yes-associated protein (YAP) has been shown to interact with and to enhance p73-dependent apoptosis in response to DNA damage. Here, we show that YAP requires the promyelocytic leukemia gene (PML) and nuclear body localization to coactivate p73. YAP imparts selectivity to p73 by promoting the activation of a subset of p53 and/or p73 target promoters. Endogenous p73, YAP, and p300 proteins are concomitantly recruited onto the regulatory regions of the apoptotic target gene *p53AIP1* only when cells are exposed to apoptotic conditions. Silencing of YAP by specific siRNA impairs p300 recruitment and reduces histone acetylation on the *p53AIP1* target gene, resulting in delayed or reduced apoptosis mediated by p73. We also found that YAP contributes to the DNA damage-induced accumulation of p73 and potentiates the p300-mediated acetylation of p73. Altogether, our findings identify YAP as a key determinant of p73 gene targeting in response to DNA damage.

Introduction

Large protein complexes that include transcription factors and specific coactivators frequently govern ac-

tivation of specific sets of genes (Naar et al., 2001). YAP, the first protein in which a WW domain was identified, is a phosphoprotein of 65 kDa that interacts with the SH3 domain of the *c-src* protooncogene product, a nonreceptor tyrosine kinase of the Src family (Sudol, 1994). WW domains are protein-protein interaction modules that recognize short proline-rich motifs of diverse proteins involved in various signaling pathways (Sudol and Hunter, 2000). The name refers to two signature tryptophan (WW) residues that are spaced 20–22 amino acids apart and play an important role in the domain structure and function (Sudol and Hunter, 2000). In addition to a type I WW domain, YAP also contains a PDZ interaction motif, an SH3 binding motif, and a coiled-coil domain. A recent work has reported the identification of a 14-3-3 binding molecule, named TAZ (transcriptional coactivator with PDZ binding motif), that shares a remarkable homology with YAP (Kanai et al., 2000). It has recently been shown that YAP is a potent transcriptional coactivator. YAP binds to and coactivates the Runx and the four TEAD/TEF transcription factors (Yagi et al., 1999; Vassilev et al., 2001).

A close link between YAP and the transcription factors of the p53 family has recently emerged (Strano et al., 2001; Basu et al., 2003). We had originally reported that YAP engages in a physical association with p73 and p63, the two newly discovered p53 family members (Strano et al., 2001). Unlike p53, p73 and p63 undergo alternative splicing, giving rise to a family of proteins whose individual function is still to be elucidated (Yang and McKeon, 2000; Levrero et al., 2000; Blandino and Dobbela, 2004). In terms of binding to p53 family members, YAP possesses two levels of specificity, binding to long, but not to short, forms of p73 and p63, and it does not bind to p53 at all (Strano et al., 2001). The binding of YAP to p73 or p63 results in a strong transcriptional coactivation (Strano et al., 2001; Basu et al., 2003; S.S. and G.B., unpublished data). A recent work has reported that YAP is phosphorylated by AKT, and such modification impairs YAP-nuclear translocation and attenuates p73-mediated apoptosis (Basu et al., 2003).

Here, we report that p73 is required for the nuclear translocation of endogenous YAP in cells exposed to cisplatin and that YAP is recruited by PML into the nuclear bodies (NBs) to promote p73 transcriptional activity. We found that YAP contributes to p73 stabilization in response to DNA damage and promotes p73-dependent apoptosis through the specific and selective coactivation of apoptotic p73 target genes and potentiation of p300-mediated acetylation of p73. Indeed, endogenous p73, YAP, and p300 proteins are concomitantly recruited to the regulatory regions of the apoptotic target gene *p53AIP1* only when cells are exposed to apoptotic conditions. Silencing of YAP by specific siRNAs impairs p300 recruitment onto the p73 apoptotic target gene *p53AIP1* upon DNA damage, and this correlates with the reduction of histone acetylation at the same promoter site and delayed or reduced apoptosis. Altogether, our results identify YAP as an impor-

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tant determinant for p73 target gene specificity through p300 recruitment and p73 acetylation.

Results

YAP Is a Selective Coactivator of p73-Dependent Transcription

YAP has been shown to act as a transcriptional coactivator of the p53 paralog p73 (Strano et al., 2001) and to potentiate p73-dependent apoptosis (Basu et al., 2003). We sought to investigate how YAP promotes p73 α transcriptional activation to increase p73-mediated apoptosis in response to DNA damage. By employing transient transactivation assays we found that YAP imparts selectivity to p73 by promoting the activation of a subset of p53 and/or p73 target promoters. Indeed, as shown in Figures 1A–1C, YAP enhanced p73 α -mediated activation of the apoptotic target genes *BAX* and *p53AIP1*, and this activation was dose dependent. The same effect was not observed when the cell cycle arrest *p21^{waf1}* gene was targeted (Figures 1D and 1E). Interestingly enough, the transcriptional activation of *BAX* by p73 γ , a shorter p73 isoform that does not bind YAP (Strano et al., 2001) or wild-type p53, was not coactivated upon YAP coexpression (Figure 1A). The thymidine kinase (TK) promoter, which does not contain any p53 and/or p73 consensus, was neither activated by p73 α nor coactivated by YAP (Figure 1F).

YAP Requires PML to Coactivate p73-Dependent Transcription

Acetylation of the p73 protein by the p300 acetyltransferase has been shown to be a major determinant of p73 proapoptotic activity in response to DNA damage (Costanzo et al., 2002). Because PML has been recently shown to bind p73 in the NBs to promote p300-mediated acetylation of p73 and to regulate its transcriptional activity (Bernassola et al., 2004), we aimed to assess whether PML contributes to the selectivity of YAP-mediated transcriptional coactivation of p73. We found that YAP is unable to cooperate with p73 α in the transcriptional activation of apoptotic target genes in *PML*^{−/−} cells (Figures 1G and 1H, and data not shown). Reconstitution of *PML*^{−/−} MEFs with wild-type PML, but not with a PML mutant unable to be properly incorporated into NBs (Fogal et al., 2000), fully restores YAP's ability to coactivate p73 α (Figure 1I). These observations imply that YAP might be recruited into the NBs to act as a transcriptional coactivator of p73 α . Indeed, confocal microscopy experiments reveal that exogenously expressed wild-type PML and GFP-tagged YAP colocalize in the NBs (Figure 1L, row b). Interestingly, the NBs' localization of YAP is lost when GFP-YAP is coexpressed with a PML mutant that cannot be sumoylated (PML IV SUMO [−]) and is therefore unable to localize into the NBs (Figure 1L, row c). These results indicate that YAP is recruited by PML into the NBs and thereby promotes p73 transcriptional activity.

YAP Enhances p73-Dependent Cell Death in Response to DNA Damage

To further investigate the biological output of YAP coactivation, we engineered a stable cell line (H-p73 α /

YAP-GFP) in which the inducible p73 expression is coupled with the constitutive expression of GFP-tagged YAP (Figure 2A, top). The control cell line H-p73 α /GFP expressed comparable amounts of the 27 kDa GFP protein (data not shown). Reciprocal coprecipitation experiments and pull-down assays revealed the formation of protein complexes containing p73 α and GFP-tagged YAP in the H-p73 α /YAP-GFP cells upon ponasterone A (Pon A) stimulation (data not shown). Both endogenous YAP (data not shown) and exogenously expressed GFP-tagged YAP (Figure 2A, bottom) displayed a cytoplasmic distribution in untreated cells. Upon Pon A induction of p73 α expression, YAP mostly localized in the nucleus (Figure 2A, bottom). GFP localization remained unchanged upon addition of Pon A to the cell culture medium of the H-p73 α /GFP cells (Figure 2A, bottom), thus demonstrating the specificity of p73-mediated YAP relocalization. It has been previously shown that Akt-mediated phosphorylation of YAP impairs its nuclear translocation and attenuates p73-mediated apoptosis (Basu et al., 2003). Accordingly we found that inhibition of Akt activity potentiates the nuclear relocalization of GFP-tagged YAP induced by p73 expression (Figure 2A, bottom).

We next treated H-p73 α /YAP-GFP and H-p73 α /GFP cells with cisplatin (CDDP, 2.5 μ g/ml), doxorubicin (Doxo, 2 μ M), and Etoposide (Eto, 5 μ M). p73 α expression increased CDDP- (Figure 2B) and Doxo-mediated (see Figure S1A in the Supplemental Data available with this article online) cell death of H-p73 α /GFP cells. A further increase was documented when YAP and p73 α were coexpressed as in H-p73 α /YAP-GFP cells (Figure 2B and Figure S1A). Unlike CDDP and Doxo, Eto-mediated cell death was not augmented by p73 α or YAP expression (Figures S1B and S1C). These findings indicate that YAP displays a high degree of specificity in the potentiation of p73-mediated apoptosis.

YAP Is Recruited In Vivo onto p73 Apoptotic Target Genes upon DNA Damage

To analyze the mechanisms underlying the selectivity of YAP coactivation of p73-dependent transcription in vivo, we first performed chromatin immunoprecipitation experiments in the H-p73 α /GFP-YAP cells under conditions that recapitulate cell cycle arrest and/or apoptosis in response to p73 accumulation (Pon A treatment) and activation (Pon A plus exposure to CDDP) (Fontemaggi et al., 2002 and Figures 2C–2E). Crosslinked chromatin derived from untreated and CDDP-treated H-p73 α /GFP-YAP cells, in the presence or absence of Pon A-induced p73 overexpression, was immunoprecipitated with a mixture of anti-p73 α polyclonal sera or with anti-YAP polyclonal serum, respectively. We found that p73 was recruited on the regulatory regions of the apoptotic target genes *p53AIP1* and *BAX* only under apoptotic conditions, i.e., Pon A plus CDDP (Figures 2Ca and 2Cb, lanes 9–12). Differently from *p53AIP1* and *BAX*, the p53 and/or p73 target genes *PIG3* and *APAF-1* did not display any p73 binding under the same conditions (Figures 2Cc and 2Cd, lanes 9–12). p73 was found on the *p21^{waf1}* promoter both in cells treated with Pon A and CDDP alone as well as in cells treated with both compounds (Figure 2Ce, lanes 9–12). This latter obser-

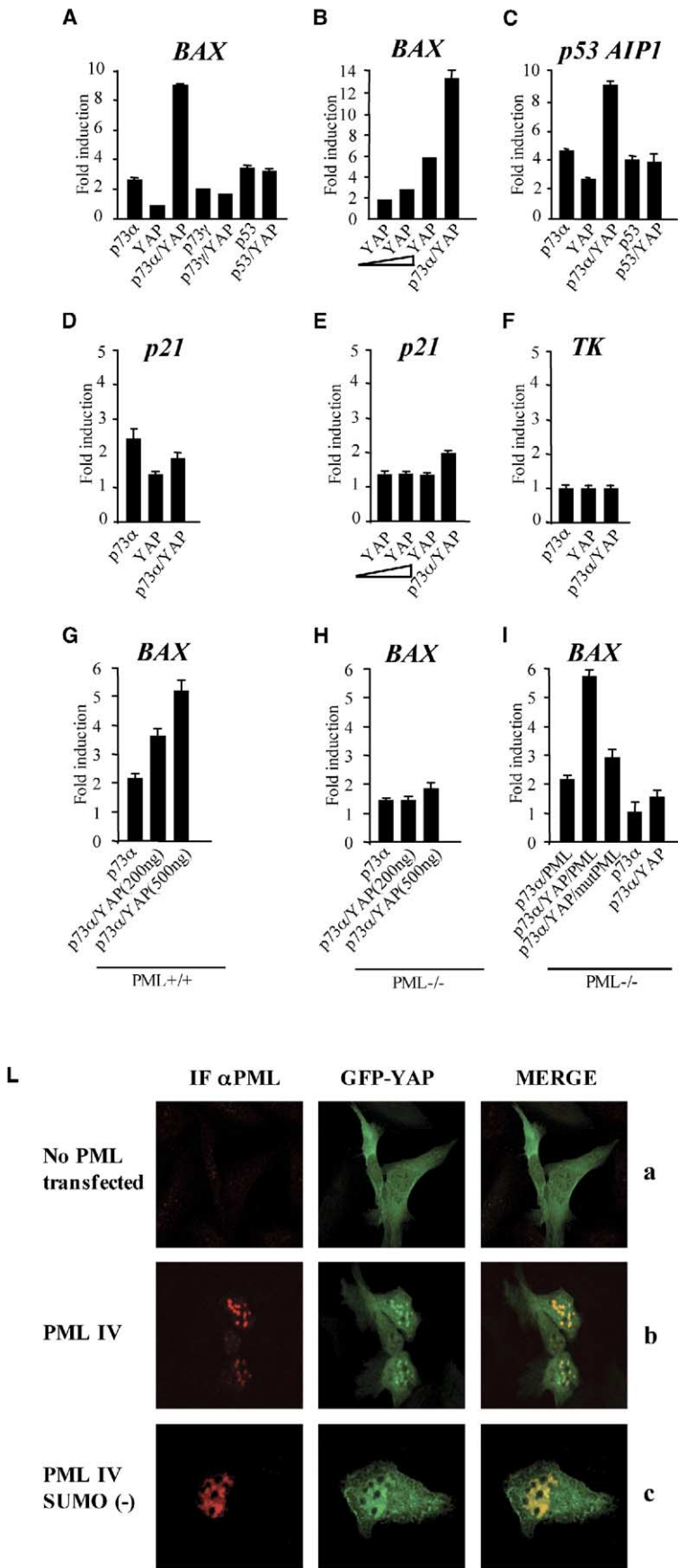


Figure 1. YAP Is a Selective Transcriptional Coactivator of p73

(A–F) H1299 cells were transiently transfected with indicated combinations of plasmids encoding p73α (A–F), p73γ (A), or p53 (A and C) (25 ng/60-mm dish), and full-length human YAP (100 ng/dish) (A–F) together with *BAX* (A and B), *p53AIP1* (C), *p21^{waf1}* (D–E), and *TK* (F) luciferase reporter plasmids (50 ng/dish). Bars indicate SD.

(B and E) Increasing amounts (100, 200, and 500 ng/dish) of YAP plasmid were transfected with the indicated luciferase reporter plasmids. Bars indicate SD.

(G) *PML^{+/+}* and (H and I) *PML^{-/-}* MEFs were transiently transfected with indicated combinations of plasmids encoding p73α (25 ng/dish) (G–I), full-length human YAP (200–500 ng/dish) (G–I), wild-type PML (1 μg/dish) (I), and SUMO (–) mutant PML (1 μg/dish) (I) together with *BAX* (G–I) luciferase reporter plasmids (50 ng/dish). The total amount of transfected DNA in each dish was kept constant by the addition of empty vector wherever necessary. Transfection efficiency normalization was performed by cotransfecting a β-gal activity expression vector and assessing β-gal activity in cell lysates. Cell extracts were prepared 36 hr later and subjected to determination of luciferase activity. Results are represented as fold induction over the control. Histograms show the mean of three experiments, each performed in quadruplicate. Bars indicate SD.

(L) U2OS cells seeded on glass coverslips were transfected with pEGFP-YAP together with either pcDNA3-PMLIV (row b), pcDNA3-PMLIV SUMO (–) encoding a variant of PMLIV that cannot be sumoylated due to mutations in all the three known sumoylation sites (Fogal et al. 2000) (row c), or with empty pcDNA3 plasmid (row a). Cells were then fixed and sequentially incubated with polyclonal anti-PML antiserum (PG-M3, Santa Cruz) and TRITC-conjugated anti-rabbit secondary antibody (Sigma). Images were analyzed by confocal scanning laser microscopy with a Zeiss Axiovert 100 M microscope attached to a LSM 510 confocal unit.

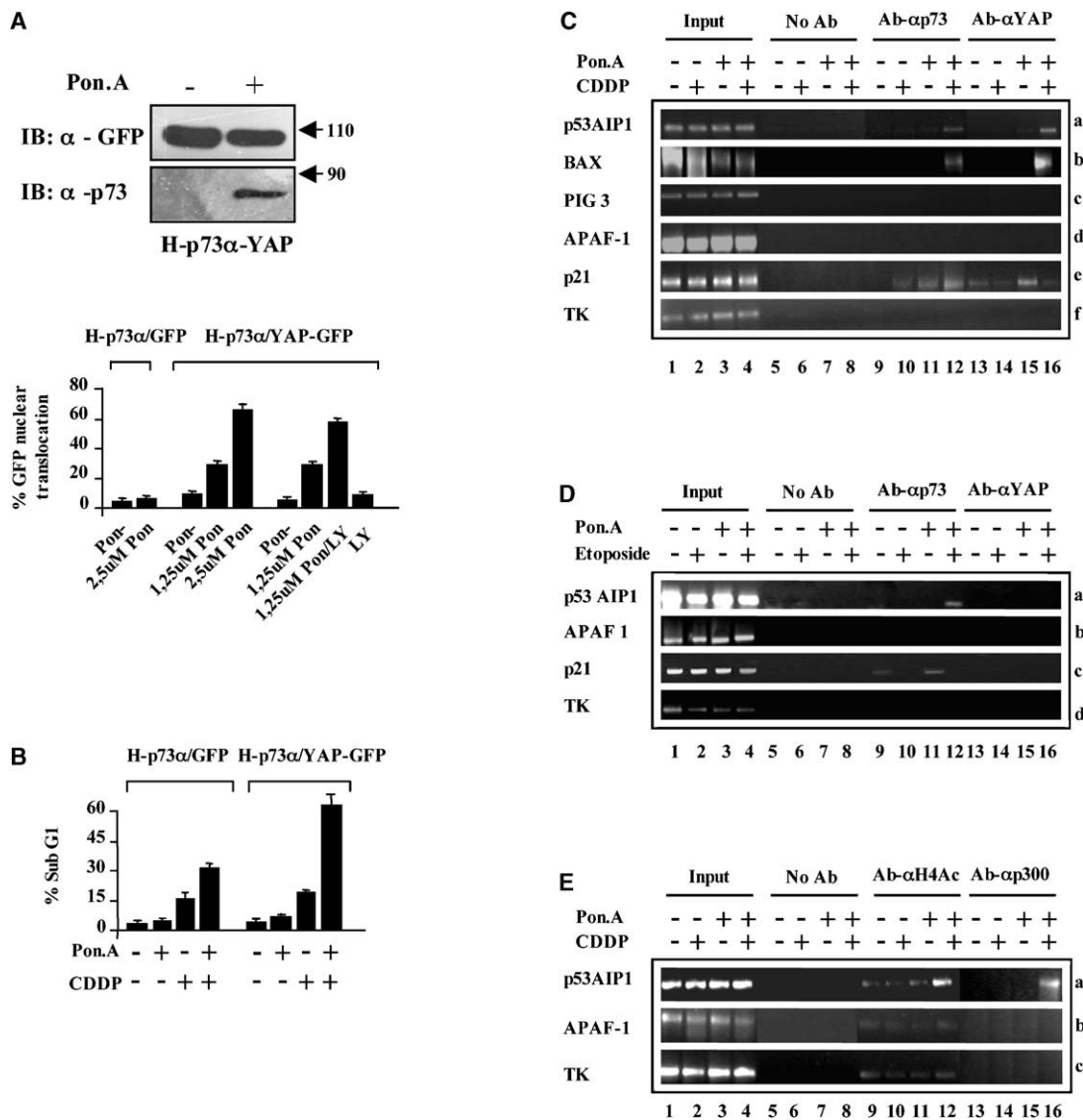


Figure 2. YAP Is Recruited In Vivo onto p73 Apoptotic Target Genes in Response to DNA Damage

(A) Subcellular localization of GFP-tagged YAP is modulated by p73 expression. H-p73α/GFP and H-p73α/YAP-GFP cells were fixed and stained with Hoechst either in absence or 24 hr after the addition of Pon A (2.5 μM) to the culture medium; aliquots containing 100 μg of total cell protein were subjected to immunoblotting with a monoclonal anti-p73 antibody and anti-GFP polyclonal serum (top); histograms in the lower panel represent the mean of three experiments in which the number of cytoplasmic and nuclear GFP-stained cells were counted. Cell clones expressing different levels of both GFP-YAP and GFP alone were employed. LY294002 (Calbiochem) at the concentration of 25 μM was added for 6 hr to the H-p73α/YAP-GFP cells pretreated with the suboptimal concentration of 1.25 μM Pon A for 12 hr. Bars indicate SD.

(B) YAP potentiates p73α-mediated apoptosis. H-p73α/GFP and H-p73α/YAP-GFP cells were treated as indicated in the [Experimental Procedures](#) section. Floating and adherent cells were collected 48 hr after addition of cisplatin, fixed in cold methanol, washed in PBS, and resuspended in 1 ml PBS containing 50 μg/ml RNase and 50 mg/ml propidium iodide. The content of DNA was measured by an Epics XL analyzer. Percentages of sub-G1 fractions are shown. Histograms represent the mean of three experiments. Bars indicate SD.

(C and D) Crosslinked chromatin derived from H-p73α/YAP-GFP cells treated with CDDP (C) or Eto (D) was immunoprecipitated with antibodies to p73 or YAP or in the absence of antibody and analyzed by PCR with specific primers for the indicated regulatory regions. Input: nonimmunoprecipitated crosslinked chromatin.

(E) Crosslinked chromatin derived from H-p73α/YAP-GFP treated as in (C) was immunoprecipitated with antiacetylated histone H4 or p300 antibodies or in the absence of antibody and analyzed by PCR with specific primers for the indicated regulatory regions. Input: nonimmunoprecipitated crosslinked chromatin.

vation is in good agreement with the known cell cycle-arrest response of these cells to p73 induction (Fontemaggi et al., 2002). Notably, the analysis of YAP chromatin immunoprecipitates has revealed that YAP is indeed

recruited in vivo onto p73 target genes (Figure 2C). We found that YAP presence on the regulatory regions of *p53AIP1* and *BAX* paralleled that of p73 and occurred only under apoptotic conditions (i.e., Pon A plus CDDP)

(Figures 2Ca and 2Cb, lanes 13–16). Similar to p73, YAP was not recruited onto *PIG3* and *APAF-1* target genes under the same conditions (Figures 2Cc and 2Cd, lanes 13–16). No specific occupancy by p73 and YAP was found on the TK promoter, which does not contain any p73 binding sites (Figure 2Cf, lanes 9–16). More importantly, the amount of YAP recruited onto the *p21^{waf1}* promoter was significantly reduced when cells were exposed to both Pon A and CDDP (Figure 2Ce, lanes 13–16). Thus, in the case of the *p21^{waf1}* promoter, differently from the apoptotic target genes *BAX* and *p53AIP1*, the binding of p73 and YAP is dissociated, strongly supporting a role for YAP as a selective coactivator of p73 proapoptotic genes. Because we showed that YAP did not increase Eto-dependent apoptosis in p73 α -expressing cells (Figures S1C and S1D) (whereas, in the same cell line, YAP strongly potentiated CDDP- and Doxo-induced apoptosis upon p73 expression [Figure 2B and Figure S1B]), we assessed the pattern of p73 and YAP recruitment onto p53 and/or p73 target genes in response to Eto treatment. p73 was recruited onto the *p53AIP1* gene under apoptotic conditions (Pon A plus Eto) and onto the *p21^{waf1}* under growth-arrest conditions (Pon A without Eto). Interestingly, YAP was never recruited, either on the *p53AIP1* or on the *p21^{waf1}* promoter, in cells exposed to Eto (Figure 2Dc, lanes 13–16). Taken together these results confirm that Eto-induced apoptosis is YAP independent. The functional meaning of p73 recruitment without YAP on the *p53AIP1* promoter remains to be investigated. These results indicate that YAP is recruited onto p53 and/or p73 target genes in vivo. To assess whether the binding of YAP to chromatin was direct or involved the formation of protein complexes with p73, we performed gel retardation assays with a labeled oligonucleotide derived from the *p21^{waf1}* promoter. In vitro-translated p73 β protein displayed a specific binding to the *p21^{waf1}* promoter that was competed with by cold oligonucleotide and abrogated by p73 antibody. No specific binding of YAP was found under the same conditions, indicating that YAP does not bind directly to p53 and/or p73 target genes (Figure S2).

We next verified whether the recruitment of YAP on p73 target genes favors chromatin modifications associated with activation of transcription. To this end, crosslinked chromatin derived from H-p73 α /GFP-YAP cells was immunoprecipitated with either a mixture of antibodies that recognize the p300 acetyl transferase (Naar et al., 2001) or antiacetylated H4 histone antibodies. As shown in Figure 2Ea (lanes 13–16), p300 was bound onto *p53AIP1* regulatory region only under apoptotic conditions, when both p73 and YAP were recruited and H4 histone acetylation was strongly induced (Figure 2Ea, lanes 9–12). Different from *p53AIP1*, neither p300 occupancy nor the increased acetylation of H4 was revealed on *APAF-1* regulatory regions (Figure 2Eb, lanes 9–16), in agreement with previous observations indicating that in response to DNA damage, *APAF-1* is mostly activated by p53 (Pediconi et al., 2003).

Altogether our data show that YAP acts as a selective transcriptional coactivator of p73 to activate its apoptotic target genes in response to DNA damage. Furthermore, we show that under apoptotic conditions,

p73, YAP, and p300 are selectively recruited to activate transcription from the regulatory region of the apoptotic *p53AIP1* target gene.

Endogenous p73, YAP, and p300 Proteins Are Concomitantly Recruited onto the *p53AIP1* Target Gene in Response to CDDP and Doxorubicin

p73 has been shown to play an important role in CDDP-induced apoptosis of HCT116(3) cells (Gong et al., 1999; Costanzo et al., 2002). We found that endogenous YAP and p73 proteins form readily detectable protein complexes in vivo in HCT116(3) exposed to apoptotic DNA damage (Figures 3A and 3B). Under the same conditions, endogenous YAP efficiently translocates into the nucleus of HCT116(3) cells (Figure 3C). Next, to investigate the behavior of endogenous YAP as a transcriptional coactivator of p73 target genes in vivo, we performed chromatin immunoprecipitation experiments in HCT116(3) cells treated with apoptotic dosages of either CDDP or Doxo. p73 was found on the *p53AIP1* gene regulatory sequences in untreated cells, and it was further recruited in treated cells (Figure 3D). Notably, YAP was found on *p53AIP1* only upon CDDP and Doxo treatment, whereas no occupancy by p73 and YAP was observed on the *p21^{waf1}* promoter under the apoptotic conditions used in these experiments (Figure 3D). We next analyzed whether the presence of both p73 and YAP on the *p53AIP1* regulatory region is associated with transcriptional activation. We found that the recruitment of p300 onto the *p53AIP1* gene under apoptotic conditions was coupled to a marked increase in H4 histone acetylation (Figure 3D). The increase of H4 histone acetylation on the *p21^{waf1}* promoter in HCT116(3) cells treated with either CDDP or Doxo was not related to the recruitment of either p73 or its coactivators YAP and p300 (Figure 3D). Finally, CDDP- and Doxo-induced p73, YAP, and p300 concomitant recruitment onto the *p53AIP1* gene was associated with the induction of *p53AIP1* transcript accumulation (Figure 3E).

Silencing of YAP Expression Reduces p73-Mediated Apoptosis

To further define the contribution of YAP to CDDP- and Doxo-mediated cell death of HCT116(3) cells, two unrelated YAP-specific small interfering RNA (siRNA) oligonucleotides were used. We found that abrogation of YAP expression remarkably reduced cell death induced by apoptotic doses of CDDP and Doxo (Figure 4A and Figure 4B, bottom), causing a significant delay/reduction in PARP cleavage (Figure 4B, top) and abolishing the transcriptional activation of the endogenous *p53AIP1* gene in response to CDDP (Figure 4C, top). Removal of endogenous YAP does not result in any modulation of endogenous *p21^{waf1}* and *BAX* (Figure 4C, middle and bottom). The effect of two unrelated, specific p73 siRNA oligonucleotides on CDDP- and Doxo-induced apoptosis in HCT116(3) cells was comparable to that observed with the YAP siRNA (Figure 4A), thus suggesting, albeit not proving, that most of p73-mediated apoptosis involves YAP activity. No change in HCT116(3) cell viability was observed when unrelated

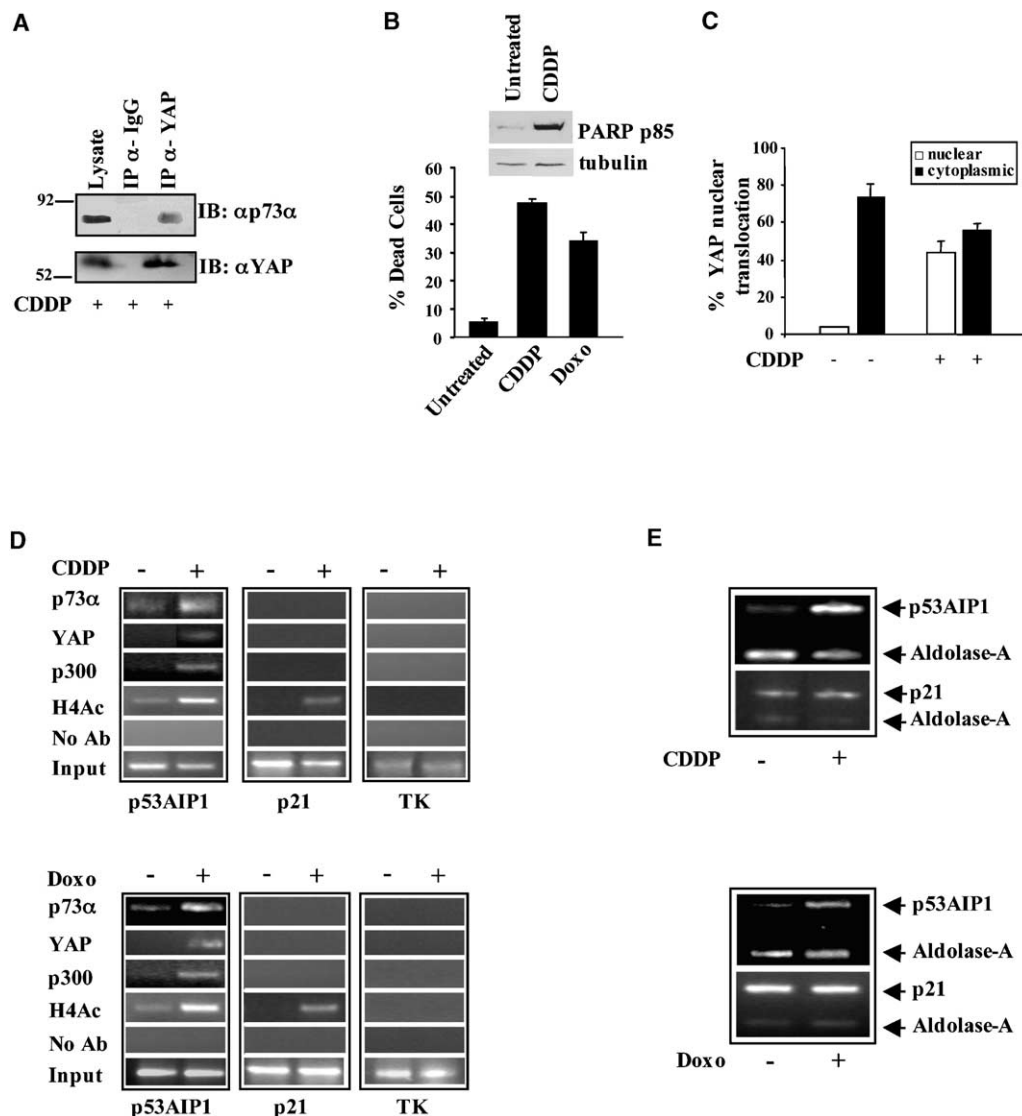


Figure 3. Endogenous p73 α , YAP, and p300 Proteins Are Concomitantly Recruited onto the Regulatory Region of *p53AIP1* in Response to CDDP and Doxorubicin

(A) Coimmunoprecipitation between endogenous YAP and p73 proteins in CDDP-treated HCT116(3) cells. Cell lysates were immunoprecipitated with anti-YAP or anti-IgG polyclonal sera, blotted onto nitrocellulose, and probed with anti-p73 (top row) and anti-YAP (top row) antibodies. Positions of protein molecular size markers are indicated on the left.

(B) Floating and adherent HCT116(3) cells treated as indicated were collected and counted with a Thoma's hemocytometer. Cell viability was determined by the ability to exclude trypan blue. Histograms represent the mean of three experiments. PARP cleavage in CDDP-treated HCT116(3) cells is analyzed by immunoblotting (top). Bars indicate SD.

(C) CDDP treatment induces nuclear translocation of YAP. HCT116(3) cells treated with CDDP as in (A) and (B) were stained with anti-YAP polyclonal serum as described in the [Experimental Procedures](#). Histograms represent the mean of three experiments in which the number of cytoplasmic and nuclear YAP-stained cells was counted. Bars indicate SD.

(D) Crosslinked chromatin derived from HCT116(3) cells treated as in (B) was immunoprecipitated with anti-p73, anti-YAP, anti-p300, anti-acetylated histone H4, or in absence of antibody and analyzed by PCR for the regulatory regions of *p53AIP1*, *p21^{waf1}*, and *TK*.

(E) RNA preparations derived from HCT116(3) cells treated as in (B) were subjected to RT-PCR analysis. Specific primers for the detection of *p53AIP1* (top) and *p21^{waf1}* (bottom) transcripts were used. Amplification of aldolase A (Ald-A) was used to normalize equal loading of each RNA preparation.

(GFPi) siRNA oligonucleotides were used (data not shown).

YAP Is Required for p300 Recruitment onto the *p53AIP1* Target Gene and Potentiates p300-Mediated Acetylation of p73

To provide a mechanistic insight on how YAP collaborates with p73 specifically at the level of its proapo-

ptotic target genes, crosslinked chromatin derived from HCT116(3) treated with YAP or with GFP siRNA oligonucleotides was immunoprecipitated with anti-p300 and anti-acetylated H4 antibodies. As shown in [Figure 5A](#), the CDDP-induced recruitment of p300 onto the *p53AIP1* target gene is markedly impaired when YAP expression is abrogated by specific siRNA. The observed reduction of p300 recruitment is paralleled by a

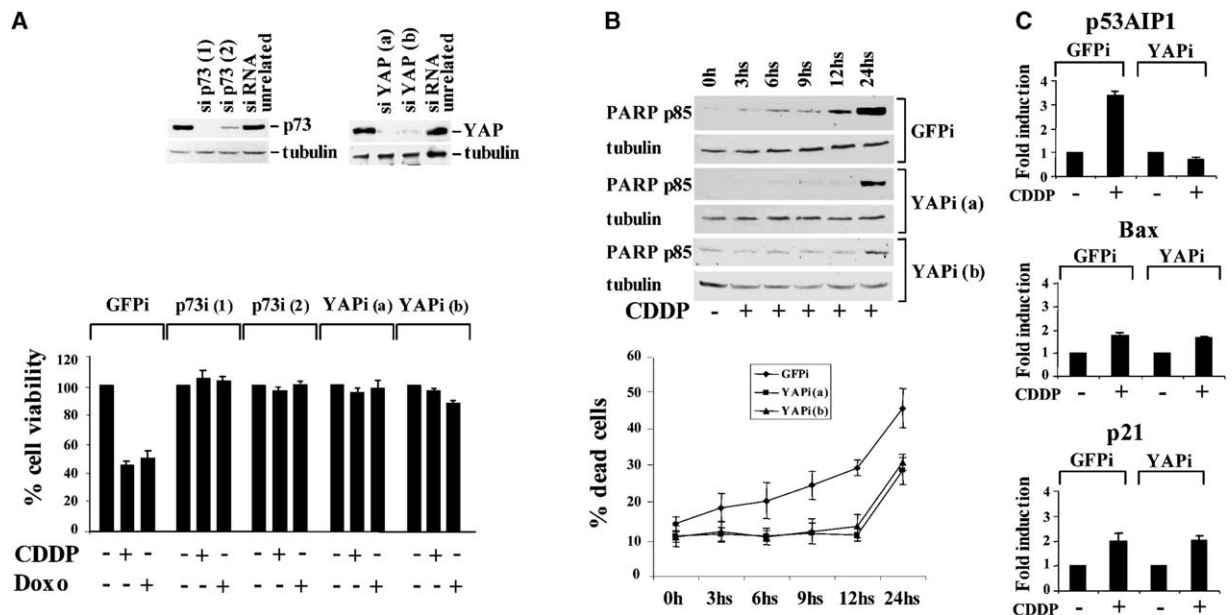


Figure 4. Silencing of YAP Decreases p300 Recruitment onto p53 AIP1 Target Gene and Reduces p73-Mediated Apoptosis

(A) Specific anti-YAP, anti-p73, and control anti-GFP siRNAs were transfected into HCT116(3) cells exposed to apoptotic dosages of either CDDP or Doxo. Cell viability was assessed by trypan blue exclusion. The efficiency of siRNA-mediated reduction of YAP and p73 expression was assessed by immunoblotting (top). Bars indicate SD.

(B) Time course of PARP cleavage (top) and cell death (bottom) in HCT116(3) cells exposed to 7.5 μ g/ml of CDDP. Bars indicate SD.

(C) RNA preparations derived from HCT116(3) cells treated as in (A) were subjected to RT-PCR analysis. Specific primers for the detection of *p53AIP1* (top), *BAX* (middle), and *p21^{waf1}* (bottom) transcripts were used. Amplification of *Ald-A* was used to normalize equal loading of each RNA preparation. Bars indicate SD.

marked decrease of H4 histone acetylation (Figure 5B). Activated Akt phosphorylates YAP on serine 127 and blocks YAP nuclear translocation (Basu et al., 2003). We found that the expression of a constitutively active Akt mutant (Bellacosa et al., 1998) abrogates the in vivo recruitment of YAP onto the *p53AIP1* target gene in cells exposed to CDDP (Figure 5C), thus providing direct evidence that AKT modulates p73-dependent apoptosis controlling the recruitment of YAP onto p73 apoptotic target genes. Because acetylation of p73 by the p300 acetyltransferase plays a major role in regulating p73 proapoptotic activity and p73 gene target selectivity (Costanzo et al., 2002), we investigated whether YAP influences p300-p73 α complex formation. As shown in Figure 5D, YAP expression favors the interaction between p73 α and p300 in cells treated with CDDP and potentiates p300-mediated acetylation of p73 (Figure 5D). Interestingly, the p73(KKK) mutant, which cannot be acetylated by p300 and is not recruited onto the *p53AIP1* apoptotic target gene in response to DNA damage (Costanzo et al., 2002), binds YAP as efficiently as wild-type p73 in a GST pull-down assay (Figure 5E). This observation is consistent with a model in which the binding of YAP to p73 precedes its acetylation and the formation of the transcriptionally active p300-p73 complex onto the chromatin.

YAP Contributes to p73 Accumulation in Response to DNA Damage

p73 accumulates in response to a number of DNA damaging agents as a result of both transcriptional activation (Pediconi et al., 2003) and increased protein sta-

bility (Agami et al., 1999; Gong et al., 1999; Costanzo et al., 2002; Mantovani et al., 2004). However, the molecular determinants of p73 protein stability upon DNA damage are not yet fully defined. We found that silencing of YAP by specific siRNAs impairs p73 accumulation in cells exposed to CDDP (Figure 6A). Notably, the p73 Y487P mutant, which is unable to bind YAP and cannot be transcriptionally coactivated (Strano et al., 2001), does not accumulate upon overexpression of YAP (Figure 5B, bottom) and displays a reduced half-life when compared to wild-type p73 α (Figure 5C). Unlike cAbl, Pin1, and p300, which promiscuously modulate p53 family members' protein stability, our findings implicate YAP as a specific interactor in the control of p73 accumulation in response to DNA damage.

YAP Requires p73 to Promote Cisplatin-Induced Cell Death

YAP activity as a transcriptional coactivator is well established, but most of the evidence supporting this notion comes from overexpression experiments (Strano et al., 2001; Basu et al., 2003). Whereas Akt-mediated phosphorylation contributes to the cytoplasmic localization of YAP to attenuate the apoptosis induced by exogenously expressed p73 (Basu et al., 2003), Pon A-induced p73 expression leads to massive YAP nuclear localization (Figure 2B). As far as endogenous YAP is concerned, we showed that its localization is almost exclusively cytoplasmic, and it translocates into the nucleus upon CDDP treatment in HCT116 (3) cells (Figure 3C). To further analyze how YAP and p73 cooperate to induce apoptosis in cells exposed to DNA

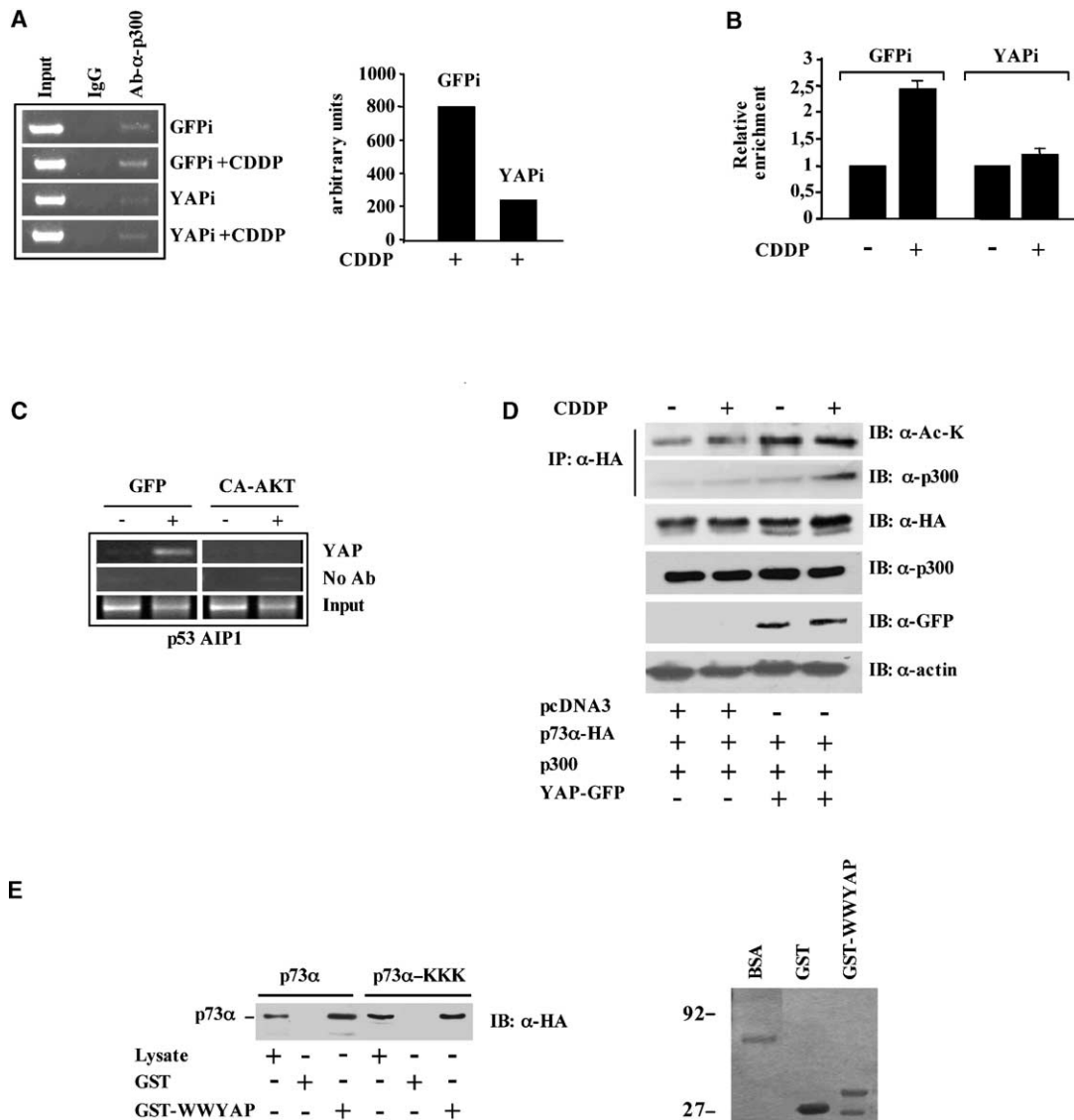


Figure 5. YAP Potentiates p73-p300 Functional Interactions in Response to DNA Damage

(A and B) Recruitment of p300 (A) and histone H4 acetylation (B) onto the *p53AIP1* target gene after CDDP treatment is decreased in HCT116(3) cells transfected with YAP-specific siRNA. In (B), the amount of *p53AIP1* or *TK* DNAs present in immunoprecipitations from HCT116(3) cells transfected with siRNAs directed against YAP or GFP (control) and treated or not with CDDP was calculated relative to the input (as described in the [Experimental Procedures](#)). Bars indicate SD.

(C) AKT controls YAP recruitment onto p73 apoptotic target genes. HCT116(3) cells were transfected with a plasmid encoding the T308D/S473D double mutant of Akt that is constitutively active (CA-AKT) ([Bellacosa et al., 1998](#)). Constitutive phosphorylation of YAP by CA-AKT impairs its *in vivo* recruitment onto the *p53AIP1* promoter in cells exposed to CDDP.

(D) Expression of YAP favors the binding of the acetyltransferase p300 to p73 α and its acetylation in response to DNA damage. Cell lysates from untreated and CDDP-treated HCT116(3) cells transfected with the indicated combinations of expression vectors were immunoprecipitated with an anti-HA monoclonal antibody and probed with antiacetylated lysines, anti-p300, and anti-p73-specific antibodies, respectively. Bottom: aliquots containing 100 μ g of total cell proteins were subjected to immunoblotting with anti-p300, anti-GFP, and anti-actin antibodies, respectively.

(E) YAP efficiently binds nonacetylatable p73. H1299 cells were transiently transfected with either wild-type p73 α or mutant p73 α (KKK) ([Costanzo et al., 2002](#)) expression vectors. Cell extracts were assayed by GST pull down with either purified GST-WWYAP or GST alone.

damage, we compared endogenous YAP localization and its recruitment onto p73 proapoptotic target genes in p73 $^{-/-}$ fibroblasts with their p73 β -reconstituted counterparts. We first found that DNA damage-induced nuclear translocation of endogenous YAP is severely impaired in cells lacking all the p73 isoforms ([Figure 7A](#)). Furthermore, when crosslinked chromatin derived

from p73 $^{-/-}$ and p73 β fibroblasts was immunoprecipitated with anti-YAP antibody, we found that endogenous YAP was selectively recruited onto the regulatory region of the proapoptotic p53 and/or p73 target gene *Killer/DR5* only in p73 β reconstituted cells upon CDDP treatment ([Figure 7B](#)). YAP was not found on the *p21^{waf1}* and the unrelated *TK* promoters under the

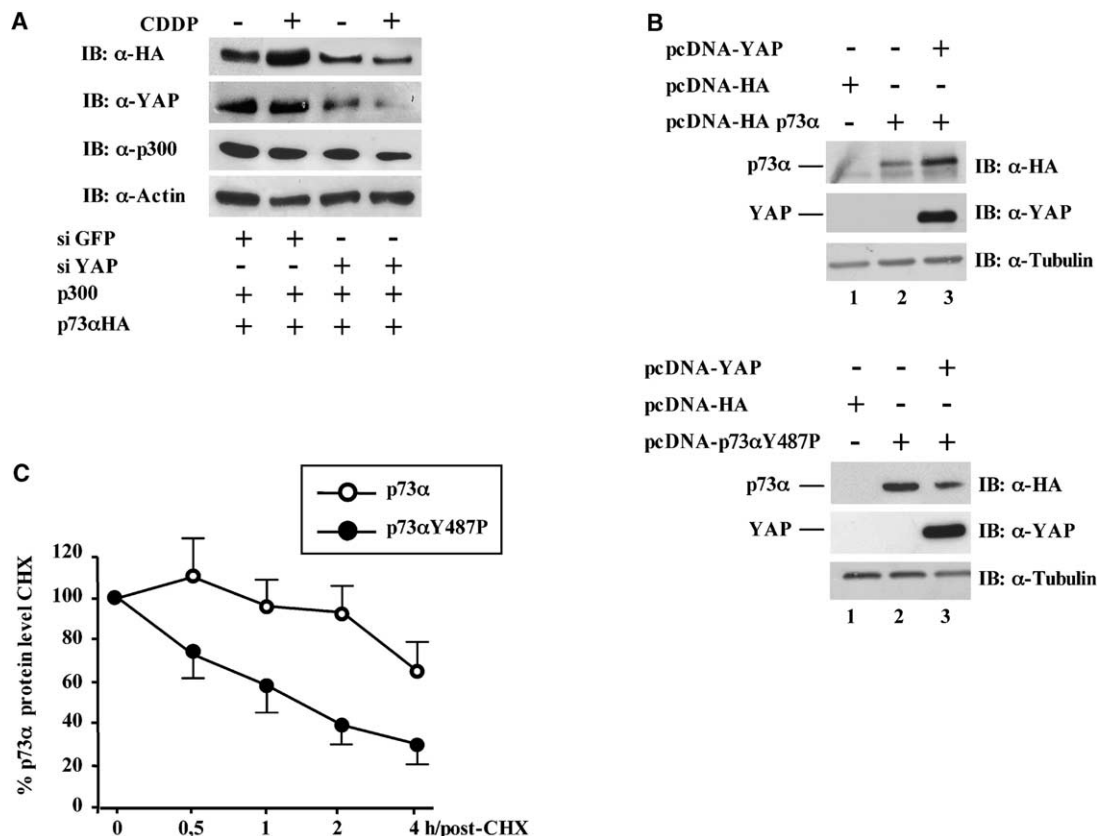


Figure 6. YAP Contributes to p73 Accumulation in Response to DNA Damage

(A) Silencing of YAP by specific siRNAs impairs p73 accumulation in cells exposed to CDDP. HCT116(3) cells were transiently transfected with p300 (6 μ g) and p73 α HA (3 μ g) expression vectors and either the specific YAP(b) siRNA or with the control GFP siRNA. Cell lysates were analyzed by immunoblotting with the indicated antibodies.

(B) The p73 Y487P mutant, which is unable to bind YAP and cannot be transcriptionally coactivated (Strano et al., 2001), does not accumulate upon overexpression of YAP. H1299 cells were transiently transfected with the pcDNA-YAP (full-length human YAP) (4 μ g) and either the pcDNA-73 α HA (2 μ g) or the pcDNA-73 α HA Y487P (2 μ g) plasmids. Cell lysates were analyzed by immunoblotting with the indicated antibodies.

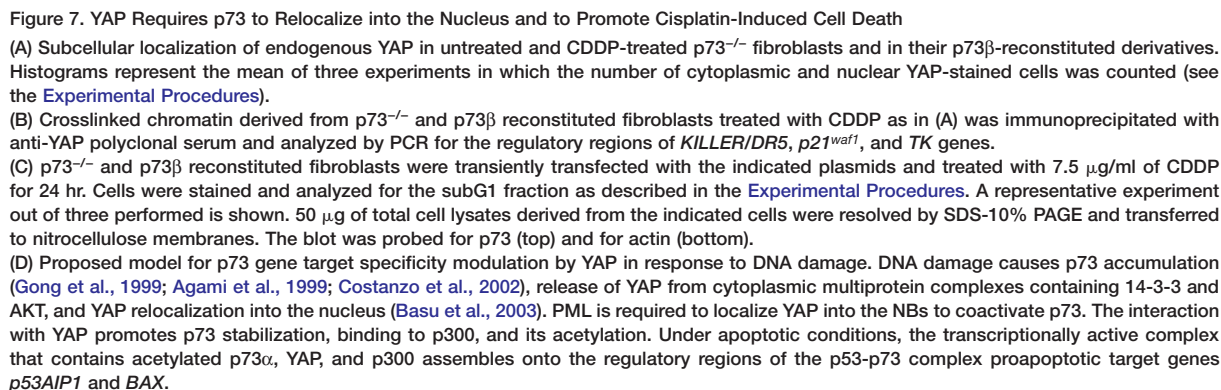
(C) The p73 Y487P mutant displays a reduced half-life when compared to wild-type p73 α . H1299 cells were transiently transfected with either the pcDNA-p73 α HA (1 μ g) or the pcDNA-p73 α HA Y487P (1 μ g) plasmids. After 24 hr, the cells were treated with 20 mg/ml of cycloheximide (CHX) for 2 hr to block de novo protein synthesis. Cell lysates, prepared at the indicated times, were processed by immunoblotting with anti-HA antibodies and quantified by scanning densitometry. The results of three independent experiments (mean $\pm$ SEM [Figure 5C] are shown).

same experimental conditions (Figure 7B). Finally, transiently overexpressed GFP-tagged YAP potentiates CDDP-mediated apoptosis only in p73 β -expressing cells (Figure 7C). Taken together, these results define a critical and mutual requirement of YAP and p73 in p73-mediated apoptotic response to DNA damage. In cells exposed to DNA damage, p73 is required for endogenous YAP to translocate into the nucleus, where it promotes p73 acetylation by p300 and dictates the selective recruitment of p73 and p300 onto p73 proapoptotic target genes.

Discussion

In this paper, we provide insights into the molecular mechanisms that regulate p73 transcriptional selectivity in the apoptotic response to DNA damaging drugs. Despite the recent identification of specific p73-target genes by genome-wide expression profile techniques (Fontemaggi et al., 2002), p73-mediated apoptosis oc-

curs mostly through the activation of a set of genes that were originally found to be activated by p53 (Costanzo et al., 2002; Irwin et al., 2003; Bergamaschi et al., 2003). This implies that selectivity at the transcriptional level is imposed to p73-mediated apoptosis by upstream events such as posttranslational modifications and specific protein-protein interactions (Costanzo et al., 2002; Agami et al., 1999; Gong et al., 1999; Yuan et al., 1999; Strano et al., 2001). An important cofactor involved in the full activation of p73 in response to DNA damage and other stressors is the prolyl isomerase Pin1 (Mantovani et al., 2004). However, Pin1 activity is also involved in DNA damage-induced activation of p53 (Zacchi et al., 2002; Zheng et al., 2002) and cannot be a determinant in dictating the final biological outcome of the cell response to DNA damage, i.e., growth arrest versus apoptosis. Selective activation of specific subsets of target genes is increasingly recognized as a crucial event in the choice between p73-mediated growth arrest or apoptosis in response to DNA damage (Co-



stanzo et al., 2002; Bergamaschi et al., 2003). Here, we show that the transcriptional coactivator YAP (Sudol and Hunter, 2000) determines p73 gene targeting in response to DNA damage. Endogenous p73, YAP, and p300 protein are concomitantly recruited to the regulatory regions of the p73 apoptotic target gene *p53AIP1* only when cells are exposed to apoptotic conditions, resulting in histone acetylation and transcriptional activation. By combining silencing of YAP expression with specific siRNAs and chromatin immunoprecipitation experiments to assess p73 target promoter occupancy, we showed that abrogation of YAP expression impairs p300 recruitment onto the *p53AIP1* target gene, and this correlates with the reduction of histone acetylation and delayed or reduced apoptosis. We also found that YAP requires PML to coactivate p73 and provided the evidence that YAP localizes into the NBs to exert its function as a specific coactivator of p73. To gain further mechanistic insight into how YAP determines p73 transcriptional selectivity, we analyzed the impact of YAP on p73 acetylation and stability—key events for the recruitment of its apoptotic potential (Agami et al., 1999; Gong et al., 1999; Costanzo et al., 2002; Flores et al., 2002). Indeed, YAP expression favors p300-p73 protein complex formation and potentiates p300-mediated acetylation of p73. The ability of YAP to efficiently bind the nonacetylatable p73 mutant p73(KKK) (Costanzo et al., 2002) suggests that the interaction between YAP and p73 precedes its acetylation and the formation of the transcriptionally active p300-p73 complex. We also show that silencing of YAP markedly impairs p73 accumulation in response to DNA damage. Our results identify YAP as a critical player for p300-p73 interactions and further link p300-mediated acetylation to p73 stabilization in response to DNA damage. On the other hand, experiments performed in p73^{-/-} and p73^β-reconstituted cells indicate that p73 is required for YAP nuclear translocation in response to DNA damage and for YAP recruitment at the DNA level to promote cisplatin-induced cell death. Altogether our results support a model of mutual cooperation between YAP and p73 in p73-mediated apoptosis: in cells exposed to DNA damage, p73 relocalizes endogenous YAP into the nucleus, where it is required to promote p73 acetylation by p300 and the selective recruitment of p73 and p300 onto p73 proapoptotic target genes.

Interestingly, YAP binds only to long p73 and p63 isoforms and not to p53, p73, and p63 short isoforms. Thus, interaction with YAP seems to regulate the fine-tuning of DNA damage-induced apoptosis by p53 family members. Indeed, lack of YAP recruitment might account for the lower efficacy of p73 short isoforms in inducing antitumoral effects (S.S. and G.B., unpublished data). The observation that YAP potentiates p73 activity only in response to specific anticancer treatments and that no recruitment of YAP onto apoptotic target genes was observed during p73-independent apoptosis indicates that YAP contributes, in addition to p73 posttranslational modifications, to define the selectivity of p73-mediated apoptosis.

Experimental Procedures

Cell Culture and Transfections

Human lung carcinoma cell line H1299 and human colon carcinoma cell line HCT116(3) were maintained in RPMI and DMEM supple-

mented with 10% fetal calf serum (FCS), respectively. The H1299-p73^α inducible cell line was generated as previously described (Fontemaggi et al., 2002). To induce expression of p73^α, Pon A, a synthetic analog of ecdysone, was added to the medium (final concentration, 2.5 μM). H-p73^α/YAP-GFP and H-p73^α/GFP were generated from the H1299-p73^α inducible cell line stably transfected with YAP-GFP and GFP expression vectors, respectively. p73^{-/-} and p73^β reconstituted mouse embryo fibroblasts (kindly provided by Dr. J.Y. Wang), *PML*^{-/-}, and +/+ MEFs (kindly provided by Dr. S. Soddu) were maintained in DMEM supplemented with 10% FCS. Transient transfections were performed either by using the calcium phosphate precipitation method (BES) or with Lipofectamine Plus reagent (GIBCO-BRL).

Recombinant Proteins and In Vivo Binding Assays

A pull-down assay was performed with 20 μg of immobilized purified GST-p73^α fusion protein or wild-type GST as previously described (Strano et al., 2001).

Immunoprecipitation

Immunoprecipitations were performed as previously described (Strano et al., 2001) with the following antibodies: anti-p73 monoclonal antibody (Ab-4, Neomarker), anti-HA (12CA5, Boehringer), or (F7AC, Santa Cruz) monoclonal antibodies and anti-YAP polyclonal serum. The latest antibodies were raised by immunizing Sprague-Dawley rats weighing 250 grams intramuscularly with naked plasmid DNA encoding full-length human YAP (kindly provided by Dr. Sudol). Hypotonic DNA solution (2 mg/ml in 0.03 M PBS [pH 7.2]) was delivered by needle syringe. Animals were injected weekly with 0.1 mg of plasmid (50 μl) for three weeks. All procedures involving animals and their care followed institutional guidelines in compliance with national and international laws and policies. Immuno-complexes were precipitated with protein G agarose beads (KPL, Guilford, CA). For p73 detection, a p73 monoclonal antibody was used at a 1:200 dilution, and for GFP detection an anti-GFP polyclonal antibody (Invitrogen) was used at a 1:5000 dilution.

Immunoblot Analysis

Total cell lysates were prepared as previously described (Strano et al., 2001). The following antibodies were used: p73 monoclonal antibody (Ab-4, Neomarker) at a 1:200 dilution, an anti-GFP polyclonal antibody (Invitrogen) at a 1:5000 dilution, a polyclonal anti-HA antibody (Y11, Santa Cruz) at a 1:250 dilution, a polyclonal anti-p300 antibody (N15, Santa Cruz) at a 1:250 dilution, an anti-acetylated lysines polyclonal antibody (catalog number 9441, Cell Signal) at a 1:1000 dilution, and an anti-PARP p85 fragment polyclonal antibody from Promega at a 1:400 dilution.

Indirect Immunofluorescence and Confocal

Microscopy Analysis

Untreated and CDDP-treated HCT116(3), p73^{-/-}, and p73^β cells were fixed and processed as already described in Strano et al. (2001). The anti-YAP polyclonal serum was used at a 1:300 dilution for 2 hr. For the confocal microscopy experiments, U2OS cells were fixed and subjected to immunofluorescence analysis with polyclonal anti-PML antibody (PG-M3, Santa Cruz) and TRITC-conjugated anti-rabbit secondary antibody (Sigma) as described (Fogal et al., 2000). Images were analyzed by confocal scanning laser microscopy with a Zeiss Axiovert 100M microscope attached to a LSM 510 confocal unit.

Cell Cycle Analysis

H-p73^α/YAP-GFP and H-p73^α/GFP cells were collected, processed, and analyzed as already described (Fontemaggi et al., 2001). p73^{-/-} fibroblasts and p73^β reconstituted with p73^β expression plasmid were transfected with the Lipofectamine Plus Reagent (Invitrogen) according to the manufacturer's protocol, with 4 μg of GFP expressing vector or the same amount of YAP-expressing vector. 16–18 hr after transfection, cells were either left untreated or were treated with 7.5 μg/ml CDDP (TEVA) for 24 hr. Cells were then washed with 1× PBS and resuspended in 1× PBS/Triton X-100, 0.1 %/PI 0.1 mg/ml, and the percentages of apoptotic cells were evaluated with an Epics-XL analyzer. Data were analyzed with the CellFIT program.

Luciferase Assays

Transient reporter assays were performed as described (Strano et al., 2002). Control plasmid (pCMVneo) was added to equally normalize the amount of transfected DNA for each transfection. The following luciferase reporter plasmids were used: Bax-luc and p21^{waf1}-luc plasmids (from Dr. Oren) and p53AIP1-luc (from Dr. Costanzo).

RNA Extraction and Reverse Transcriptase Reaction

RNAs were isolated and reverse transcribed as previously described (Fontemaggi et al., 2002). Polymerase chain reaction (PCR) analyses were performed by using oligonucleotides specific for the p53AIP1, p21^{waf1}, BAX, and the housekeeping aldolase A genes (available upon request).

Formaldehyde Crosslinking and Chromatin Immunoprecipitation

Chromatin immunoprecipitation experiments were performed as previously described (Fontemaggi et al., 2002). The following antibodies were used: a mixture of anti-p73 polyclonal antibodies (sc-7237 and sc-7238, Santa Cruz), anti-acetyl-histone H4 (catalog number 06-866, Upstate Biotechnology, Inc.), a mixture of anti-p300 polyclonal antibodies (sc-584 and sc-585, Santa Cruz), and rat polyclonal anti-YAP. Levels of p53AIP1, BAX, PIG3, p21^{waf1}, APAF1, KILLER/DR5, and TK DNAs were determined by PCR with oligonucleotides spanning the p73 binding sites. The sequences of the oligonucleotides used are available upon request.

The LightCycler system (Roche Applied Science) was used for quantitative PCR analysis of the ChIP assay shown in Figure 5B. Amplified DNA was detected with LightCycler FastStart DNA Master SYBR Green I (Roche Applied Science) by following the manufacturer's instructions. The amount of p53AIP1 or TK sequence present in samples immunoprecipitated with anti-acetylated histone H4 (H4Ac) or with no antibody (negative control) was calculated relative to a standard curve obtained by using serial dilutions of the input sample derived from each chromatin. The value of the no-antibody sample was subtracted from that obtained with the anti-H4Ac, as a nonspecific binding background, to generate corrected values that were used to calculate folds of induction between CDDP-treated and nontreated samples. Relative enrichment represents the ratio between folds of induction obtained for p53AIP1 and TK (control gene) (Nicolas et al., 2003; Kaeser and Iggo, 2002). Figure 5B shows the result of three independent real-time PCRs. For p53AIP1 and TK amplifications, the same oligonucleotides described above were used.

siRNA Studies

HCT 116.3 cells were transfected with the Lipofectamine Plus Reagent (Invitrogen) according to the manufacturer's protocol with 1 μ g of GFP expressing vector and with either 2 μ g of the control small interfering RNA siGFP or with siYAP RNA, or with sip73 RNA. 5 hr after transfection, cells were either treated with 2 μ M Doxorubicin (Sigma) or with 7.5 μ g/ml CDDP (TEVA) for 24 hr or left untreated. Apoptotic cells were then evaluated by the trypan blue incorporation. siRNAs were purchased from Dharmacon Inc., and sequences are available upon request.

Supplemental Data

Supplemental Data including two figures are available online with this article at <http://www.molecule.org/cgi/content/full/18/4/447/DC1/>.

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Update

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The Transcriptional Coactivator Yes-Associated Protein Drives p73 Gene-Target Specificity in Response to DNA Damage

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In our recent article, there was an omission in the Acknowledgments section. The corrected text appears here.

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